

Current Score : 29 / 29 Due : Monday, February 26 2018 11:00 AM SAST

The due date for this assignment is past. Your work can be viewed below, but no changes can be made.

Important! Before you view the answer key, decide whether or not you plan to request an extension. Your Instructor may *not* grant you an extension if you have viewed the answer key. Automatic extensions are not granted if you have viewed the answer key.

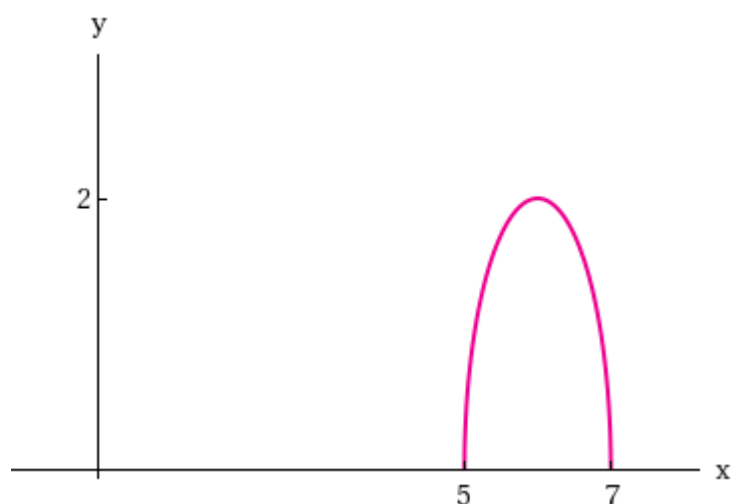
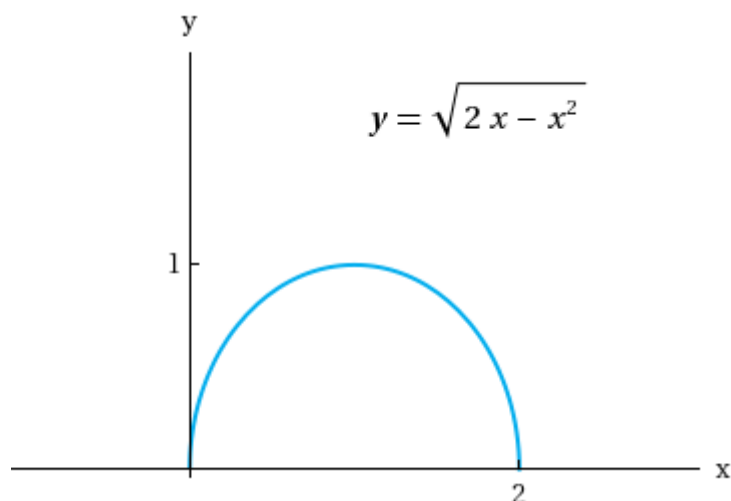
[View Key](#)

1. 3/3 points | [Previous Answers](#)SCalcET8M 1.3.006.MI.

The graph of $y = \sqrt{2x - x^2}$ is given. Use transformations to create a function whose graph is as shown.

$y =$

$2\sqrt{2(x-5)} - (x-5)^2$



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2. 2/2 points | [Previous Answers](#)SCalcET8M 1.3.043.

Express the function in the form $f \circ g$. (Use non-identity functions for f and g .)

$$F(x) = (7x + x^2)^4$$

$$\{f(x), g(x)\} = \{$$

$$x^4, (7x + x^2)$$



}

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Starting with the graph of $y = e^x$, write the equation of the graph that results from the following changes.

(a) shifting 3 units downward

$y =$

e^{x-3}



(b) shifting 9 units to the right

$y =$

e^{x-9}



(c) reflecting about the x -axis

$y =$

$-e^x$



(d) reflecting about the y -axis

$y =$

e^{-x}



(e) reflecting about the x -axis and then about the y -axis

$y =$

$-e^{-x}$



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Find each of the following functions.

$$f(x) = 1 - 4x, \quad g(x) = \cos(x)$$

(a) $f \circ g$

$$1 - 4\cos(x)$$



State the domain of the function. (Enter your answer using interval notation.)

$$(-\infty, \infty)$$



(b) $g \circ f$

$$\cos(1 - 4x)$$



State the domain of the function. (Enter your answer using interval notation.)

$$(-\infty, \infty)$$



(c) $f \circ f$

$$16x - 3$$



State the domain of the function. (Enter your answer using interval notation.)

$$(-\infty, \infty)$$



(d) $g \circ g$

$$\cos(\cos(x))$$



State the domain of the function. (Enter your answer using interval notation.)

$$(-\infty, \infty)$$



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5. 3/3 points | [Previous Answers](#)SCalcET8M 1.5.041.

Express the given quantity as a single logarithm.

$$\frac{1}{7} \ln(x + 2)^7 + \frac{1}{2} [\ln(x) - \ln(x^2 + 3x + 2)^2]$$

$\ln(\sqrt{x+1})$



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6. 2/2 points | [Previous Answers](#)SCalcET8M 1.5.017.

If $g(x) = 6 + x + e^x$, find $g^{-1}(7)$.

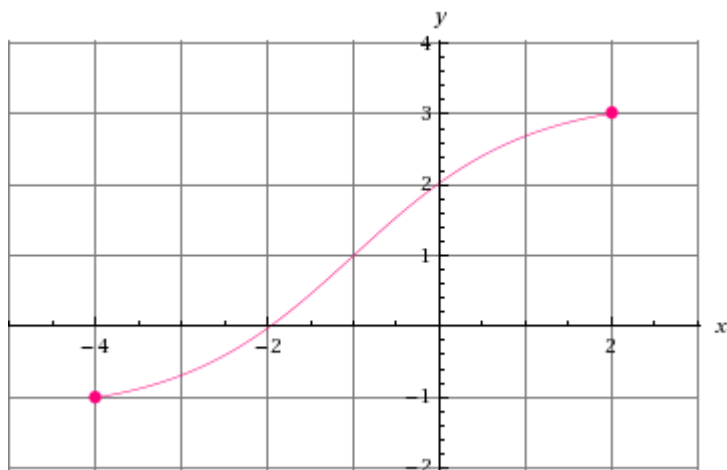
$g^{-1}(7) =$

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The graph of f is given.



(a) Why is f one-to-one?

f is one-to-one because it passes the Horizontal Line Test ✓.

(b) What are the domain and range of f^{-1} ? (Enter your answers in interval notation.)

\$\$\$[-1,3]

domain



\$\$\$[-4,2]

range



(c) What is the value of $f^{-1}(1)$?

-1



(d) Estimate the value of $f^{-1}(0)$ to the nearest tenth.

-2



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8. 1/1 points | [Previous Answers](#)SCalcET8M 1.5.024.

Find a formula for the inverse of the function.

$$y = \frac{e^x}{1 + 6e^x}$$

$$f^{-1}(x) =$$

$$\ln(1 - 6x)$$



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